MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG4DSEAQC200 –Advanced Aquaculture Production Systems

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer all 10
Each question carries 1 marks

1. Comment on organic farming. [U] [1]

2.What is meant by carrying capacity in aquaculture system [R] [1]

3. Mention one method for controlling pollution from aquaculture effluents [U] [3]

4. How does aeration affect biofloc development? [U][2]

5. What is recirculating aquaculture system? [R] [3]

6. What is bioremediation in aquaculture ? [R] [2]

7. What is the role of biofilter in aquaponics design? [U] [4]

8. Mention one advantage of flow - through culture systems ? [U] [5]

9. Comment on partitioned aquaculture system ? [U] [6]

10. What is a polyhouse aquaculture system? [R] [3]

[1x 10=20] Marks

Part B
Short Answer Questions
Answer any 5 Questions out of 8
Each question carries 6 marks

11. Describe different types of farming systems in aquaculture [R] [1]

12. Write an overview of global aquaculture production [U] [1]

13. Explain stocking practices of fin fish and shell fish in biofloc - based culture system [R] [2]

14. Elaborate suitable culture species for indoor culture system and polyhouse aquaculture [R] [3]

15. Explain water quality management and routine system maintenance in aquaponics [R] [4]

16. Describe the design and functioning of in - pond raceway system. [R] [5]

17. Explain the concept and components of integrated multitrophic aquaculture system [R] [6]

18. Discuss nutrient recycling in aquaponics system and how it helps achieve zero nutrient discharge [U] [4]

[6 x 5 = 30] Marks

Part C
Essay Type Questions
Answer Any One Question
Each question carries 10 marks

19. Compare Aquaponics, IMTA, and PAS systems with respect to nutrient recycling, resource utilisation, and environmental impact. [U] [4]

20. Discuss aquaculture waste disposal, biofiltration, and nitrifiers in controlling environmental pollution [U][3]

21. Discuss the challenges faced by global aquaculture and explain how innovative farming systems and carrying-capacity enhancement strategies can address them [An] [1]

[10 x 1 = 10] Marks